Hide menu
Module 2 Introduction
Lesson 1: Linear Regression and Least Squares
Reading: Linear Regression and Least Squares Reading 30 min
Video: What is Linear Regression? - Part 1 7 min
Video: What is Linear Regression? - Part 2 4 min
Video: Linear Regression 11 min
Video: Linear Regression Assumptions 9 min
Video: Statistical Tools 20 min
Practice Assignment: Linear Regression and Least Squares Quiz Submitted
Lab: Coding Example 1h
Lab: Coding Exercise 1h
Lesson 2: Modification of Linear Regression: Subset Selection
Lesson 3: Coefficient Shrinkage for Linear Regression: Ridge Regression and LASSO
Lesson 4: Data Transformations and Linear Regression
Module 2 Summative Assessment
Module 2 Summary

Your grade: 100%

Next item

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Linear Regression and Least Squares Quiz

Review Learning Objectives

Instructions

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

Submitted	Attempts
Nov 15, 2:34 AM IST	Unlimited

Your grade

To pass you need at least 80%. We keep your highest score.

100%

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Try again

1. In the context of linear regression, what is the meaning of a residual? **1 / 1 point**

- The proportion of variance explained by the model.
- (selected) The difference between an observed value and its corresponding predicted value based on the regression model.
- The gradient of the regression line.
- The average of all the dependent variable values.

Correct

Correct! A residual is the error term that represents the distance between the observed value and the estimated value on the regression line.

2. Which of the following is an assumption of linear regression? **1 / 1 point**

- The independent variable is measured without error.
- The relationship between the independent and dependent variable is non-linear.
- The residuals are distributed uniformly.
- (selected) The variance of the residuals is constant across all levels of the independent variable.

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Correct

3. What is the significance of the p-value in the context of the coefficients in a linear regression model? **1 / 1 point**

- It represents the proportion of the data that fits the regression model perfectly.
- (selected) It tests the null hypothesis that the coefficient is equal to zero.
- It indicates the probability that the coefficient is due to random chance.
- It is the expected value of the dependent variable when all independent variables are zero.

Correct

Correct! The p-value in linear regression tests the null hypothesis that the coefficient is zero, which would imply that there is no effect of the corresponding independent variable on the dependent variable.